

Polynomial Long Division

Answer Key

Algebra 2 Polynomial Arithmetic

Quick Answers

- | | |
|--|---|
| 1. $x + 5$ | 7. $2x^2 - 2x + 7 - \frac{10}{x+1}, -10$ |
| 2. $x + 3 + \frac{3}{x+2}$ | 8. $x^2 + 2x + \frac{2x-1}{x^2+1} - 4$ |
| 3. $x^2 - x - 6$ | 9. $x^3 + 2x^2 - x - 2$ |
| 4. $2x^2 - 3x + 4 - \frac{8}{x+3}, -8$ | 10. $3x^2 + x + 2 - \frac{2}{x-1}, 3x^3 - 2x^2 + x - 4$ |
| 5. $x^2 + 2x + 4$ | 11. $2x^2 - x + \frac{-x-2}{x^2-x+1} - 3$ |
| 6. $3x^2 + x - 2$ | 12. $x^2 - x - 2, (x-3)(x-2)(x+1)$ |
-

Curriculum

Level: Algebra 2 Polynomial ArithmeticHSA-APR.A.1 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12**Difficulty 1–5 of 5 across 16 tagged checks.*

Common wrong answers

4. *If they got 70: substituted 3 instead of the value that makes $x + 3$ zero*
7. *If they got 4: substituted 1 instead of the value that makes $x + 1$ zero*
-

1. $(x^2 + 7x + 10) \div (x + 2)$

Step 1. Leading terms: $x^2 \div x = x$. Multiply: $x(x + 2) = x^2 + 2x$. Subtract: $7x - 2x = 5x$, and bring down +10.**Step 2.** Leading terms again: $5x \div x = 5$. Multiply: $5(x + 2) = 5x + 10$. Subtract: nothing is left, so the remainder is zero.**Step 3.** With a zero remainder the answer is the quotient alone.

$x + 5$

Check: $(x + 2)(x + 5) = x^2 + 7x + 10$, the dividend.

2. $(x^2 + 5x + 9) \div (x + 2)$

Step 1. $x^2 \div x = x$; $x(x + 2) = x^2 + 2x$; subtract: $5x - 2x = 3x$; bring down +9.**Step 2.** $3x \div x = 3$; $3(x + 2) = 3x + 6$; subtract: $9 - 6 = 3$. The leftover 3 has degree 0, below the divisor's degree 1, so the algorithm stops and 3 is the remainder.**Step 3.** The remainder is written over the divisor.

$x + 3 + \frac{3}{x+2}$

Listen for: an answer written as “ $x + 3$ r3”. That is fine as scratch notation, but the form asked for puts the remainder over the divisor.

3. $(x^3 - 2x^2 - 5x + 6) \div (x - 1)$

Step 1. $x^3 \div x = x^2$; $x^2(x - 1) = x^3 - x^2$; subtract: $-2x^2 + x^2 = -x^2$; bring down $-5x$.

Step 2. $-x^2 \div x = -x$; $-x(x - 1) = -x^2 + x$; subtract: $-5x - x = -6x$; bring down $+6$.

Step 3. $-6x \div x = -6$; $-6(x - 1) = -6x + 6$; subtract: $6 - 6 = 0$. No remainder.

$$x^2 - x - 6$$

Teaching note: every subtraction flips the sign of the whole product. Missing that on the middle round is the single most common error here.

4. $(2x^3 + 3x^2 - 5x + 4) \div (x + 3)$, **with a substitution check**

Step 1. $2x^3 \div x = 2x^2$; $2x^2(x + 3) = 2x^3 + 6x^2$; subtract: $3x^2 - 6x^2 = -3x^2$; bring down $-5x$.

Step 2. $-3x^2 \div x = -3x$; $-3x(x + 3) = -3x^2 - 9x$; subtract: $-5x + 9x = 4x$; bring down $+4$.

Step 3. $4x \div x = 4$; $4(x + 3) = 4x + 12$; subtract: $4 - 12 = -8$, the remainder.

$$2x^2 - 3x + 4 - \frac{8}{x+3}$$

Step 4. The check: $x + 3$ is zero at $x = -3$, so substituting -3 into the dividend leaves only the remainder. $2(-27) + 3(9) + 15 + 4 = -54 + 27 + 15 + 4$.

$$-8$$

Common wrong answer: 70 — substituting 3 rather than the value that makes the divisor zero. The check only works at the divisor's own root.

5. $(x^3 - 8) \div (x - 2)$

Step 1. Write the dividend with every power present: $x^3 + 0x^2 + 0x - 8$. Without those placeholders the columns slip and the answer comes out a degree short.

Step 2. $x^3 \div x = x^2$; subtract $x^3 - 2x^2$ to leave $2x^2$; bring down $0x$. Then $2x^2 \div x = 2x$; subtract $2x^2 - 4x$ to leave $4x$; bring down -8 .

Step 3. $4x \div x = 4$; $4(x - 2) = 4x - 8$; subtract: 0.

$$x^2 + 2x + 4$$

Teaching note: this is the difference-of-cubes pattern $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$, found here by division rather than recalled.

6. $(6x^3 - x^2 - 5x + 2) \div (2x - 1)$

Step 1. $6x^3 \div 2x = 3x^2$; $3x^2(2x - 1) = 6x^3 - 3x^2$; subtract: $-x^2 + 3x^2 = 2x^2$; bring down $-5x$.

Step 2. $2x^2 \div 2x = x$; $x(2x - 1) = 2x^2 - x$; subtract: $-5x + x = -4x$; bring down $+2$.

Step 3. $-4x \div 2x = -2$; $-2(2x - 1) = -4x + 2$; subtract: 0.

$$3x^2 + x - 2$$

Listen for: a student who writes the first quotient term as $6x^2$, dividing only the variable parts. The coefficients divide too.

7. $(2x^3 + 5x - 3) \div (x + 1)$, **with a substitution check**

Step 1. The square term is missing, so write $2x^3 + 0x^2 + 5x - 3$ before starting.

Step 2. $2x^3 \div x = 2x^2$; subtract $2x^3 + 2x^2$ to leave $-2x^2$; bring down $5x$. Then $-2x^2 \div x = -2x$; subtract $-2x^2 - 2x$ to leave $7x$; bring down -3 .

Step 3. $7x \div x = 7$; $7(x + 1) = 7x + 7$; subtract: $-3 - 7 = -10$, the remainder.

$$2x^2 - 2x + 7 - \frac{10}{x+1}$$

Step 4. Check at $x = -1$: $2(-1) + 5(-1) - 3 = -2 - 5 - 3$.

$$-10$$

Common wrong answer: 4 — substituting 1 instead of -1 . The divisor $x + 1$ vanishes at -1 , and that is the only value the check uses.

8. $(x^4 + 2x^3 - 3x^2 + 4x - 5) \div (x^2 + 1)$

Step 1. $x^4 \div x^2 = x^2$; $x^2(x^2 + 1) = x^4 + x^2$; subtract: $-3x^2 - x^2 = -4x^2$, leaving $2x^3 - 4x^2 + 4x - 5$.

Step 2. $2x^3 \div x^2 = 2x$; $2x(x^2 + 1) = 2x^3 + 2x$; subtract: $4x - 2x = 2x$, leaving $-4x^2 + 2x - 5$.

Step 3. $-4x^2 \div x^2 = -4$; $-4(x^2 + 1) = -4x^2 - 4$; subtract: $-5 + 4 = -1$, leaving $2x - 1$. That has degree 1, below the divisor's degree 2, so the algorithm stops.

$$x^2 + 2x - 4 + \frac{2x-1}{x^2+1}$$

Teaching note: with a quadratic divisor the remainder is allowed to carry an x . Students who keep dividing until nothing is left produce a “quotient” with negative powers.

9. $(x^4 - 5x^2 + 4) \div (x - 2)$

Step 1. Fill the gaps: $x^4 + 0x^3 - 5x^2 + 0x + 4$.

Step 2. $x^4 \div x = x^3$; subtract $x^4 - 2x^3$ to leave $2x^3$. Then $2x^3 \div x = 2x^2$; subtract $2x^3 - 4x^2$ to leave $-x^2$.

Step 3. $-x^2 \div x = -x$; subtract $-x^2 + 2x$ to leave $-2x$. Then $-2x \div x = -2$; subtract $-2x + 4$ to leave 0.

$$x^3 + 2x^2 - x - 2$$

Check: the dividend factors as $(x^2 - 1)(x^2 - 4)$, so $x - 2$ really is one of its factors and a zero remainder is expected.

10. $(3x^3 - 2x^2 + x - 4) \div (x - 1)$, then multiply back

Step 1. $3x^3 \div x = 3x^2$; subtract $3x^3 - 3x^2$ to leave x^2 ; bring down x . Then $x^2 \div x = x$; subtract $x^2 - x$ to leave $2x$; bring down -4 .

Step 2. $2x \div x = 2$; $2(x - 1) = 2x - 2$; subtract: $-4 + 2 = -2$, the remainder.

$$3x^2 + x + 2 - \frac{2}{x-1}$$

Step 3. Now the check. Multiply the divisor by the quotient: $(x - 1)(3x^2 + x + 2) = 3x^3 - 2x^2 + x - 2$. Adding the remainder -2 gives the polynomial below, which is the dividend — so the division is right.

$$3x^3 - 2x^2 + x - 4$$

Why it works: division states that dividend equals divisor times quotient plus remainder. Multiplying back is not extra work, it is the definition being read backwards.

11. $(2x^4 - 3x^3 + x - 5) \div (x^2 - x + 1)$

Step 1. The square term is missing, so the dividend is $2x^4 - 3x^3 + 0x^2 + x - 5$. Then $2x^4 \div x^2 = 2x^2$; $2x^2(x^2 - x + 1) = 2x^4 - 2x^3 + 2x^2$; subtract to leave $-x^3 - 2x^2 + x - 5$.

Step 2. $-x^3 \div x^2 = -x$; $-x(x^2 - x + 1) = -x^3 + x^2 - x$; subtract to leave $-3x^2 + 2x - 5$.

Step 3. $-3x^2 \div x^2 = -3$; $-3(x^2 - x + 1) = -3x^2 + 3x - 3$; subtract to leave $-x - 2$, which has degree 1 and stops the algorithm.

$$2x^2 - x - 3 - \frac{x+2}{x^2-x+1}$$

Listen for: a sign slip in step 2. Subtracting $-x^3 + x^2 - x$ means adding $x^3 - x^2 + x$, and the two middle signs are the ones that get lost.

12. $(x^3 - 4x^2 + x + 6) \div (x - 3)$, then factor completely

Step 1. $x^3 \div x = x^2$; subtract $x^3 - 3x^2$ to leave $-x^2$; bring down x . Then $-x^2 \div x = -x$; subtract $-x^2 + 3x$ to leave $-2x$; bring down 6.

Step 2. $-2x \div x = -2$; $-2(x - 3) = -2x + 6$; subtract: 0. The quotient is the answer to the division.

Step 3. A zero remainder says $x - 3$ is a factor, so the dividend is $(x - 3)(x^2 - x - 2)$. The quotient factors further: two numbers multiplying to -2 and adding to -1 are -2 and 1.

Teaching note: this is why long division matters beyond arithmetic practice. One known factor turns an unfactorable-looking cubic into a quadratic anyone can finish.